

Ammonia decomposition in the thermochemical waste-heat recuperation systems: A view from low and high heating value

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ARTICLE INFO

Keywords:

Ammonia
Decomposition
Thermochemical recuperation
Hydrogen-rich fuel
Thermodynamic analysis

ABSTRACT

This paper is devoted to the investigation of the thermochemical waste-heat recuperation system based on ammonia decomposition. Thermochemical waste-heat recuperation system consists of a reactor, an ammonia preheater, and a condenser. The thermodynamic analysis of the TCR system was performed via Aspen HYSYS software. The heat balance was calculated for the different operating parameters such as temperature (100–700 °C) and pressure (1–20 bar). The heat balance showed that in the temperature range from 150–500 °C there are heat deficits in the TCR system. To cover this deficit up to 15% of steam from flue gas have to be condensed. For compensating the heat deficit was suggested using a condenser after a preheater. The heat of H₂O condensation is covered a significant part of the heat deficit. The heat recovery rate was determined and showed that the TCR system has a maximum efficiency in the temperature range above 500 °C and pressure above 20 bar. The heat recuperated in the reactor is 2–5 times higher than the heat recuperated in the preheater. The heat transformation coefficient (ratio between the combustion heat of the ammonia decomposition products and ammonia) is about 1.16 for in the temperature range above 500 °C.

1. Introduction

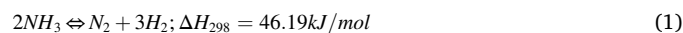
Extensive use of hydrocarbons (fossil fuels) as an energy source has resulted in the global increase of temperature and abnormal climate changes. Recent studies have pointed out that carbon dioxide and methane are the chief greenhouse gases [1]. The reduction of these greenhouse emissions and the sufficient energy supply to the energy demands let one find alternative sources of energy. Carbon-free fuels are rising up as candidates that can replace hydrocarbon fuels in the near future. Among these fuels, hydrogen occupies a special place, which has the greatest prospects to become the main fuel in the coming decades. European Union, China, United States, and other countries have programs for a carbon-free economy, where hydrogen occupies a special place [2]. Moreover, there are the perspectives of the use of other carbon-free fuels, for example, ammonia [3,4].

At present, infrastructure for a hydrogen economy is developing all over the world [5–7]. The main problems of hydrogen infrastructure are the complexity of storage and transportation of hydrogen. Due to hydrogen's low volumetric combustion heat in comparison to widely used fossil fuels, storage at high pressures is required which in turn necessitates additional energy use for its compression. In addition, hydrogen

storage in its pure form can lead to the destruction of reservoirs due to hydrogen corrosion. Therefore, at a transitional stage from a carbon economy to hydrogen economy for storage, it is proposed to use hydrogen in a bound form and, immediately before use, to extract hydrogen in a pure form [8,9].

There are various organic and inorganic hydrogen carriers, both liquid and gaseous. Liquid carriers include LOHCs (liquid organic hydrogen carrier) of various compositions [10,11]. Among gaseous carriers, ammonia (NH₃) has the greatest prospects [12,13]. Hydrogen is released from such carriers immediately before being used as a fuel.

The process of hydrogen production from ammonia is well studied [14–16]. The chemical reaction of ammonia decomposition is as follows:



As it can be seen from Eq. (1), to decompose 2 mol of ammonia and obtain 3 mol of hydrogen we need to supply 46.19 kJ of external heat. Such external heat can be obtained in a variety of ways. One of such ways is the use of waste-heat after fuel-consuming equipment. Most fuel-consuming equipment has an exhaust gas temperature of more than 100–150 °C. In this case, not only hydrogen is obtained from ammonia but also waste-heat is recovered. One part of waste-heat is spent for

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ammonia preheating and another part is supplied for a chemical reaction of ammonia decomposition. Such a way of waste-heat using is called thermochemical recuperation (TCR) [17].

An idea of waste-heat use for fuel reforming is not new. Recently, a set of papers on thermochemical waste-heat recuperation was published by various scientific groups [18–23]. Most of the works on TCR deals with the reforming of fossil fuels: natural gas [24,25], liquid fuels [26–28], coal [29,30].

Natural gas for TCR found a wide application for recuperation of the high-temperature exhaust gases after the industrial furnaces (forging, glass-melting, etc) and gas turbines. The regenerative thermochemical waste-heat recuperation system by natural gas reforming for oxy-fuel glass furnace with a capacity of 50 tons per day was realized at Pavisa (Mexico) [31,32]. The authors suggested using the methane reforming process for TCR. According to the authors of the project, the use of TCR will make it possible to reduce fuel consumption by about 30% compared to using only thermal recuperation (air heating). The specialists from GTI – Gas Technology Institute (Rue, Kozlov, Khinkis, Kurek) reported about thermochemical recuperation as a way for energy efficiency increasing of the glass furnaces [33]. The experiments and simulations showed that overall fuel savings from TCR are from 25 to 30% relative to a recuperation system based on air preheating. Popov et al. [24] presented the results of the thermodynamic analysis of different schematic diagrams of thermochemical recuperation in an industrial gas furnace. In this paper [24], the authors analyzed the TCR system based on natural gas reforming with steam (scheme A), flue gases (scheme B), and steam-flue gases mixture (scheme C). The calculations were performed for a glass furnace with a capacity of 250 tons per day at an exhaust gas temperature of 1500°C. Based on the simulation study the authors concluded that TCR by methane reforming can help to save up to 10–25% of energy without affecting other parameters of the process.

Olmsted and Grimes were the first who proposed using thermochemical recuperation based on steam methane reforming for the gas turbines [34]. The authors suggested partially replacing of steam superheater section with a steam methane reformer. The exhaust heat after a gas turbine was used for the endothermic steam methane reforming process. Various schemes of TCR implementation in the GTU were proposed in last years [35–37]. Kesser et al. performed the thermodynamic analysis of a gas turbine power plant with thermochemical recuperation via steam methane reforming [38]. The efficiency of a gas turbine with TCR was compared with a steam-injected gas turbine, a combined cycle, and a simple gas turbine without any recuperation system. The authors determined that the gas turbines with TCR with a compressor pressure ratio of 30 have a thermal efficiency of about 47.0% while for a pressure ratio of 15, the thermal efficiency is about 47.3%. Both these variants are attractively high when for a steam-injected gas turbine, a combined cycle, and a simple gas turbine without any recuperation system. Verkhivker and Kravchenko carried out the thermodynamic analysis of a combined cycle plant with thermochemical recuperation based on steam methane reforming [21]. The authors proposed using steam for reforming from a high-pressure cylinder of a steam turbine. Verkhivker and Kravchenko calculated the energy efficiency of a combined cycle plant with thermochemical recuperation has reached up to 90%.

TCR by the reforming of liquid hydrocarbons (alcohols, light oil, gasoline, etc.) has found wide application in internal combustion engines. Generally, the idea of using hydrocarbon fuel reforming to increase the energy efficiency of fuel-consuming facilities was discussed as early as the 1960s [39]. Interestingly, Lee and Wimmer initially considered pre-combustion fuel reforming as a way to reduce pollutant emissions from spark-ignition engines. And only a few years later, Martin [40] considered ICE with hydrocarbon fuel reformer as a total energy conversion system where the increase in ICE efficiency is due to the use of exhaust heat to reform the original fuel (gasoline).

A comprehensive overview of investigations on the application of

reforming of liquid fuels in internal combustion engines (ICE) was presented by Tartakovsky and Sheintuch [20]. The authors overviewed their papers on TCR by reforming of methanol, ethanol, etc, as well as other paper on this topic. Poran [41] et al. presented the results of experimental investigation of the first prototype of direct injection internal combustion engine with high-pressure thermochemical recuperation. The authors reported that such ICE demonstrated a significant improvement over the gasoline reference case with 19–30% relative increase in indicated efficiency and reduction in NO_x, CO, HC, and CO₂ emissions by up to 97, 91, 96, and 15%, respectively. A detailed thermodynamic analysis of thermochemical recuperation (TCR) applied to an idealized internal combustion engine with single-stage work extraction is presented. Results for several different fuels are included. Chakravarthy [42] et al. presented the study of the theoretical potential of thermochemical exhaust heat recuperation for internal combustion engines. They showed that for a stoichiometric mixture of methanol and air, TCR can increase the estimated ideal engine second law efficiency by about 3% for constant pressure reforming and over 5% for constant volume reforming.

Any fossil fuel can be used for thermochemical recuperation; therefore, there are a number of publications on TCR by coal gasification. Kuchonthara et al. based on the concept of thermochemical recuperative energy proposed an integrated power generation cycle combining thermochemical recuperation, brown coal gasification, and a solid oxide fuel cell (SOFC) [29]. The author's previous work is dealing with TCR by coal gasification in the industrial furnaces [30].

In our previous works, we considered various aspects of this interesting topic such as thermochemical waste-heat recuperation. For example, we developed an algorithm for calculation of the efficiency of the thermochemical waste-heat recuperation systems [43]. Moreover, we performed the thermodynamic analysis of thermochemical waste-heat recuperation by various reactions of hydrocarbon fuel reforming [44,45]. Some of our previously developed algorithms and methods have been applied in this article.

The examples above show that there are many ways to implement thermochemical waste-heat recuperation. Therefore, it can be assumed that the reaction of ammonia decomposition (1) can also be used for TCR. In this case, ammonia will be the feedstock for the production of hydrogen-rich gas, and waste-heat will be used for the ammonia decomposition reaction (1). In this regard, the purpose of this article is to study thermochemical waste-heat recuperation by ammonia decomposition, in particular, thermodynamic analysis, first law energy analysis, as well as determination of the efficiency of TCR system with and without taking into account the condensation of steam in the combustion products.

2. Thermodynamic analysis

2.1. Thermochemical recuperation concept

The main idea of thermochemical waste-heat recuperation is using waste-heat for a thermochemical transformation of initial fuel. Further, this transformed initial fuel in the new form (in the current case in the form of hydrogen-rich gas) is used for heat generation via combustion. Since the reaction of ammonia decomposition is strongly endothermic, external heat has to be supplied to the reactor. In thermochemical recuperation systems, the source for external heat is waste-heat. The schematic diagram of fuel-consuming equipment with thermochemical recuperation by ammonia decomposition is presented in Fig. 1.

The schematic diagram (Fig. 1) of the thermochemical waste-heat recuperation system is simple and consists of two main elements – reactor where the reaction of ammonia decomposition occurs; preheater where initial ammonia is slightly heated before reactor. The schematic diagram can be expanded if ammonia is used in liquid form. In this case, an ammonia regasifier has to be used. Hot exhaust gas after fuel-consuming equipment is consistently passing through reactor and

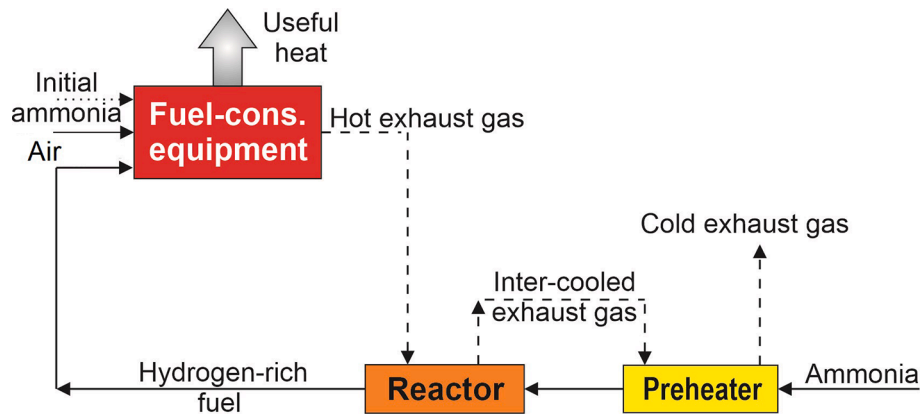


Fig. 1. The schematic diagram of fuel-consuming equipment with thermochemical recuperation by ammonia decomposition.

preheater; thereby, waste heat is recovered in the fuel cycle of fuel-consuming equipment.

The thermochemical waste-heat recuperation system is changing the process of heat extraction from ammonia. If heat extraction from ammonia via direct combustion is proceeding in one stage (Fig. 2a), then heat extraction in fuel-consuming equipment with thermochemical waste-heat recuperation is divided into two stages (Fig. 2b). The first stage is ammonia decomposition. The second stage is the combustion of hydrogen-rich fuel. It should be noted here that the combustion heat of hydrogen-rich fuel is higher than the combustion heat of initial ammonia (per kg of initial ammonia). According to the first law energy balance, the enthalpy of reaction (1) shows the difference between the combustion heat of initial ammonia and hydrogen-rich fuel.

The efficiency of the thermochemical waste-heat recuperation process is closely related to the thermodynamics of the ammonia decomposition process. To perform the first law energy analysis of thermochemical waste-heat recuperation systems as well as to determine the efficiency of recuperation, the thermodynamic analysis was performed. Moreover, the combustion products consist of 75% (vol.) of steam. Therefore, we analyzed the efficiency of the TCR system from a point of view from low and high heating values. For the last point of view, we took into account H_2O condensation.

2.2. Methodology

One of the main criteria of thermochemical recuperation efficiency is a heat recovery rate (HRR). HRR is a relation between the recuperated heat and the enthalpy of waste heat (according to Fig. 1 waste heat is presented by exhaust heat). The recuperated heat for the scheme in Fig. 1 is a sum of heat that transferred from exhaust gas to ammonia in the reactor and preheater. In other words, the recuperated heat can be determined as a sum of the enthalpy of decomposition reaction and

change of the enthalpy of ammonia by heating. Therefore, the heat recuperation rate can be determined as a following:

$$R = \frac{H_{rec}}{H_{exh}} = \frac{\Delta H_{298} + \Delta H_{am}}{H_{exh}} \quad (2)$$

where H_{rec} – a recovered heat in the TCR system; H_{exh} – the enthalpy of an exhaust gas; ΔH_{298} – an enthalpy of decomposition reaction; ΔH_{am} – the enthalpy change of ammonia.

The enthalpy of the reaction of ammonia decomposition in the expression (2) is depending on the operational parameters such as temperature and pressure. In addition, these operational parameters affect the composition of hydrogen-rich gas. Moreover, the type of catalysts is also playing an important role in the products of decomposition.

To determine the enthalpy of the decomposition reaction and hydrogen-rich gas composition as a function of the operational parameters, the thermodynamic analysis via Gibbs free energy minimization method was performed.

The thermodynamic analysis makes it possible the calculation of the equilibrium concentration of the products of ammonia decomposition at the given technological parameters such as temperature and pressure. The main concept of the Gibbs free energy minimization method is the fact the reaction system is thermodynamically favorable when its total Gibbs free energy is minimum and its differential is equal to zero [46,47]:

$$dG^t = 0 \quad (3)$$

where G^t is total Gibbs free energy.

Total Gibbs free energy of the ammonia decomposition process can be calculated as a sum of the chemical potential of all components:

$$G^t = \sum n_i \mu_i \quad (4)$$

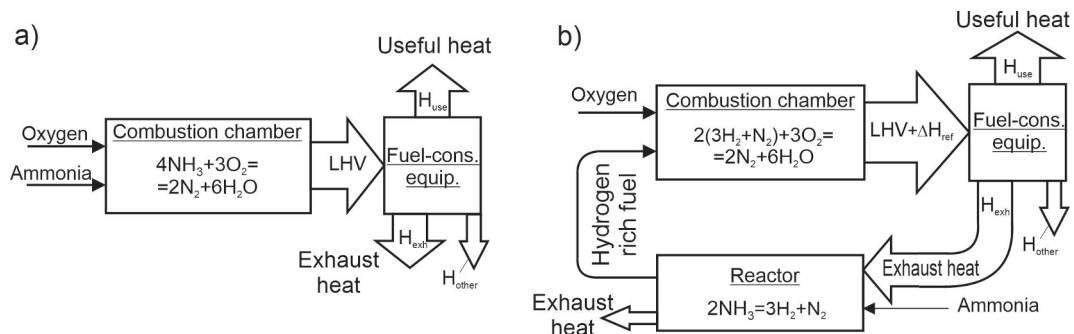


Fig. 2. The principal schemes of heat extraction from ammonia via direct combustion (a) and via thermochemical recuperation (b): LHV – lower heating value; H_i – heat fluxes.

where n_i is number of i th component moles; μ_i is chemical potential of i th component.

A chemical potential of each component can be calculated based on following expression:

$$\mu_i = \Delta G_{f_i}^0 + RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (5)$$

where $G_{f_i}^0$ is standard Gibbs free energy of i th component formation; R is molar gas constant; T is system temperature; f_i is fugacity of a pure i th component; f_i^0 is fugacity of a pure i th component at the standard state.

If we combine the equations presented above, we can write the equation for total Gibbs free energy of the chemical system:

$$G^t = \sum n_i \Delta G_{f_i}^0 = \sum n_i RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (6)$$

The thermodynamic analysis was performed via Aspen-HYSYS software that is a widely used program in chemical engineering for the modeling of chemical systems similar to the considered thermochemical recuperation system. To simulate the ammonia decomposition process, the ideal Gibbs reactor is chosen. The Gibbs reactor determines the decomposition products by the Gibbs free energy minimization method of each component based on the chemical mechanism of the ammonia decomposition process. The Peng-Robinson equation of state was used to calculate the volume of the product gas as a function of pressure and temperature [48].

The calculation scheme of fuel-consuming equipment with a thermochemical waste-heat recuperation system is presented in Fig. 3. In this scheme, a condenser is placed to take into account the heat of steam condensation.

The calculation scheme from Fig. 3 is fully repeating the schematic diagram from Fig. 1. Heat for the decomposition reaction is supplied from Recuperator1. Therefore, heat from Recuperator1 ($Q_{\text{decomposition}}$) is equal to heat supplied to Reactor. The same situation with heat for the ammonia preheating in Preheater. After Recuperator2 the exhaust gas is passed through the condenser and then the nitrogen-water mixture is fed to Separator to obtain liquid water.

The thermodynamic analysis doesn't consider the chemical kinetics, effect of catalyst, residence time, etc. on the decomposition products. The thermodynamic analysis gives us opportunity to determine the maximum (ultimate level) of the energy efficiency for the thermochemical waste-heat recuperation systems at given operation parameters such as temperature and pressure. These limit levels of the energy efficiency can be a guideline for the design of the elements of the TCR systems by ammonia decomposition.

3. Results and discussion

3.1. Equilibrium conditions

As mentioned above the thermodynamic calculations were performed with the help of Aspen HYSYS. This software is a recognized simulator for modeling the chemical process. However, in this paper, we performed verification of results to show that our results are accurate. The results of the thermodynamic analysis were compared with results of the thermodynamic analysis obtained by Ojelade and Zaman [49]. Moreover, we compared our thermodynamic results with experimental results obtained by Lucentini et al. [50]. Fig. 4 shows the comparison of the results obtained in this paper with the results of the thermodynamic analysis performed by Ojelade and Zaman [49] and with the experimental results obtained by Lucentini et al. [50].

Gibbs free energy minimization method is used for the thermodynamic analysis by Ojelade and Zaman [49]. The authors used MathLab for the simulation. Our results of the results obtained by Ojelade and Zaman are in a good correlation. A discrepancy between results is about 2% and can be explained by using different data bases of thermodynamic properties. In addition, we compared our results with the experimental results [50]. Lucentini et al. performed an experimental investigation of the ammonia decomposition process over a Ni-Ru catalysts. The experimental results from Fig. 4b were obtained over 0.6Ni-0.1Ru-CeO₂-3D catalyst at total flow 25 ml/min, $GHSV = 8300h^{-1}$, 1 bar and 10 bar. The similarity of the results is obvious.

The equilibrium composition of hydrogen-rich gas after the reactor depends on temperature and pressure. Fig. 5 presents the hydrogen-rich gas composition for various operating parameters. Fig. 5 makes it clear that product composition strongly depends on pressure and temperature. While at 200°C H₂ concentration is less than 50%, at 400°C about three-quarters of the volume is hydrogen. With regard to the problem of thermochemical waste-heat recuperation as on-board hydrogen production technology, this fact must be taken into account. Consequently, to achieve the maximum hydrogen yield in the TCR systems, it is necessary to have a heat source (for example, exhaust flue gases) with a temperature above 300–400°C. Fig. 5 shows that increasing temperature leads to increasing in hydrogen yield as well as decreasing in remain ammonia in the hydrogen-rich gas. Pressure also is playing an important role in the ammonia decomposition process. For example, at a temperature of 300°C for 1 bar the share of ammonia in the product gas is about 2% while at the same temperature for 20 bar the share of ammonia is about 20%. Therefore, pressure also has to be taken into account for the application of ammonia decomposition in the thermochemical waste-heat recuperation systems.

The efficiency of the thermochemical waste-heat recuperation

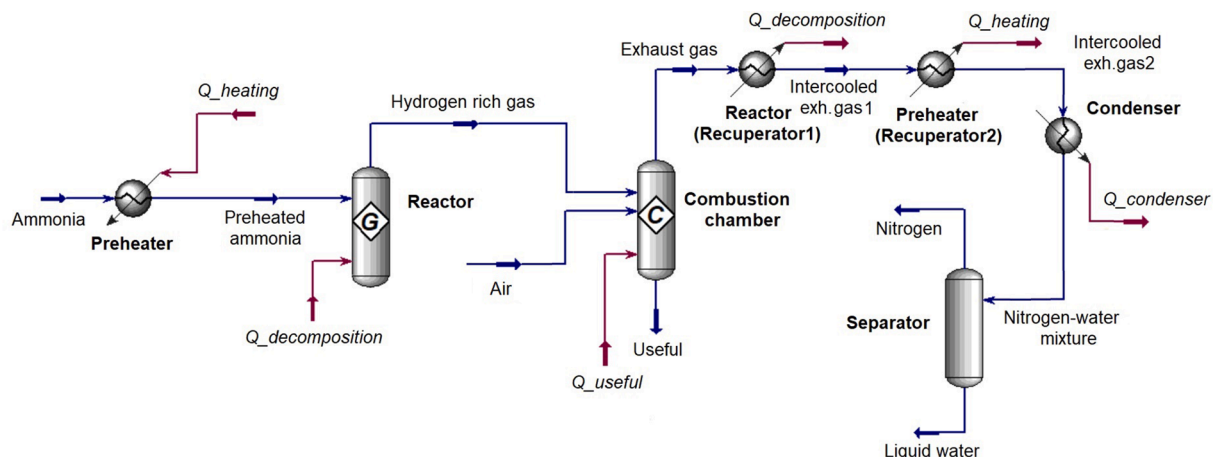


Fig. 3. The calculation scheme for thermodynamic analysis realized in Aspen HYSYS.

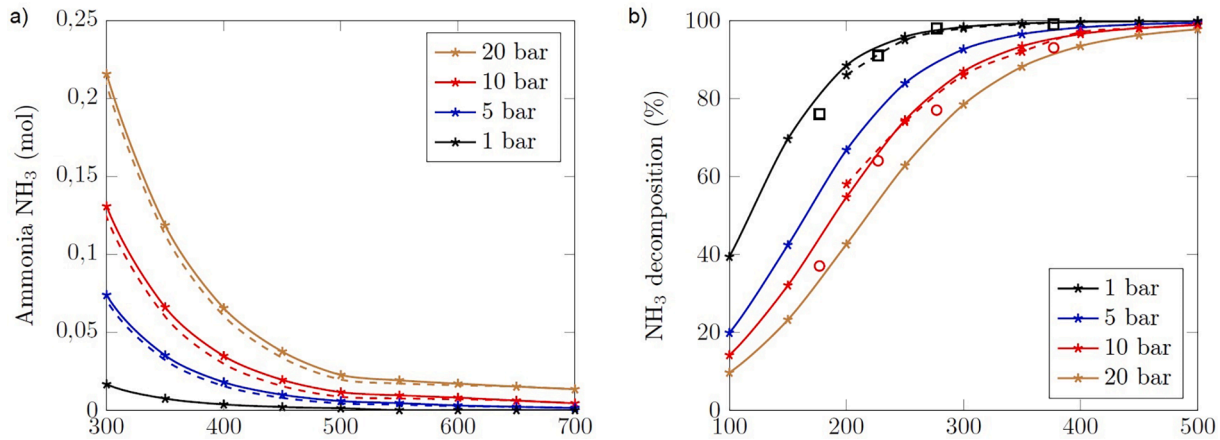


Fig. 4. Verification of the results of thermodynamic analysis: (a) Comparison with the results of thermodynamic analysis [49]; (b) Comparison with the experimental results [50] and results of thermodynamic analysis [49]. Solid lines – authors' results; dashed lines – result obtained by Ojelade and Zaman [49]; hollow points – experimental results obtained by Lucentini et al. [50].

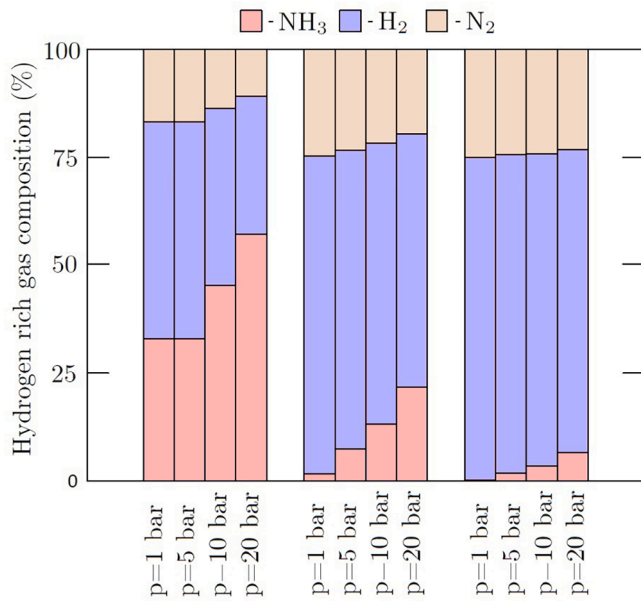


Fig. 5. The hydrogen rich gas composition after ammonia decomposition.

according to Eq. (2) depends on the enthalpy of the reaction decomposition. Fig. 6 shows the effect of the operating parameters on the enthalpy of the reaction decomposition.

Fig. 6 depicts that temperature increasing leads to an increase in the reaction enthalpy. In a temperature range from 100 to 300 °C the effect of temperature is especially noticeable. In a temperature range 300–400 °C the reaction enthalpy is slightly increasing and in the temperature range above 400 °C is reaching its maximum values. On the other hand, the pressure increasing leads to a decrease in the reaction enthalpy. For temperature 250 °C the reaction enthalpy at 1 bar is approximately twice higher than the reaction enthalpy at 20 bar. Therefore, an increase in ammonia pressure leads to a decrease in storage volume. However, on the other hand, it leads to a decrease in the reaction enthalpy.

3.2. Efficiency and heat balance

A heat recuperation rate is one on the main parameter of efficiency of the thermochemical waste-heat recuperation system. We calculated the heat balance of the TCR system when air is used as an oxidizer for

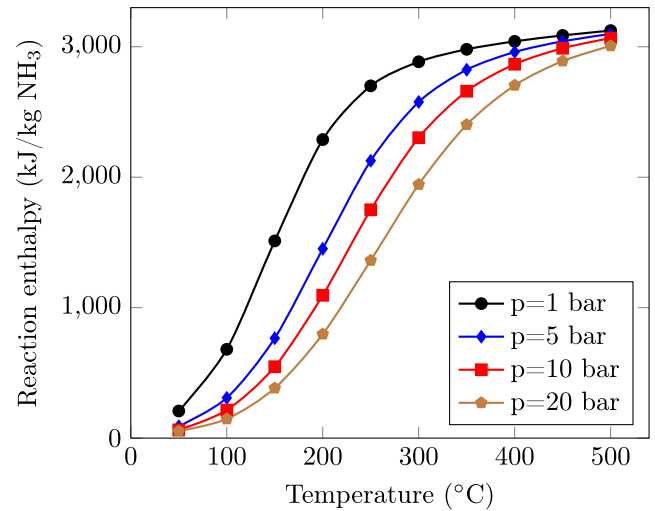


Fig. 6. The enthalpy of the ammonia decomposition reaction.

hydrogen rich gas. The air consist of about 79% of ballast nitrogen that don't react with other components but this nitrogen is a heat carrier. To calculate the heat balance we took the following assumptions: no heat losses in the thermochemical recuperation system; temperature in the reactor is 50 °C below the exhaust gas temperature; ammonia is heated in the preheater to the temperature in the reformer.

Initial conditions for the calculation:

- Mass flow rate of ammonia at inlet is 100 kg/h;
- Inlet temperature of ammonia is 30 °C;
- The excess air ratio for ammonia combustion is 1.0;
- Pressure of exhaust gas is atmospheric.

The heat balance of the thermochemical waste-heat recuperation system was calculated in the range of exhaust gas temperature from 100 to 700 °C and for the pressure of 1, and 10 bar. Fig. 7 shows the heat balance of the TCR system as a function of temperature for the pressure of 1 bar (a) and 10 bar (b).

Fig. 7 shows the heat balance of the TCR system: red line - heat flux with exhaust gas after a combustion chamber; black line - heat flux in the reactor for the decomposition process; blue line - heat flux to preheat ammonia to a temperature of the decomposition process; brown line - heat deficit; black dashed line - share of water from exhaust gas that has

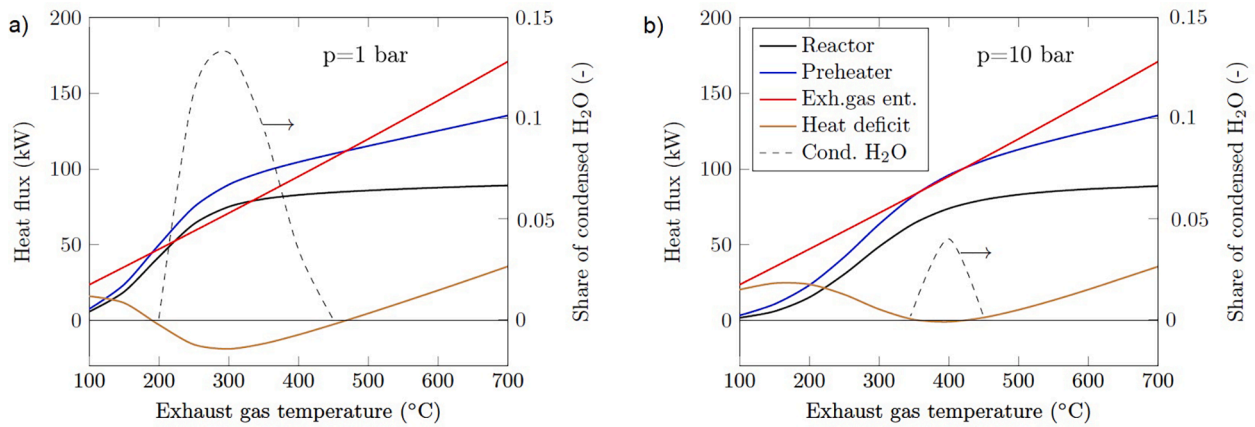


Fig. 7. The heat balance of the thermochemical waste-heat recuperation system for pressure of 1 bar (a) and 10 bar (b) and share of condensed water from exhaust gas, respectively.

to be condensed for removing the heat deficit.

As mentioned above, the temperature of the decomposition process is 50 degrees less than the temperature of exhaust gas. Fig. 7a shows that in the temperature range from 200 to 450°C is not enough exhaust heat to accomplish the ammonia decomposition process with the given initial parameters. Therefore, a heat deficit is appearing in the TCR system. The brown line just shows this deficit. A heat deficit can be compensated for by extracting heat from the combustion chamber. However, such an option will reduce the efficiency of the recuperation system. Another option for compensating the heat deficit is the use of a condenser after a preheater (Fig. 1). In the condenser, the exhaust gas will be further cooled, and the heat of water condensation from exhaust gases will be usefully used. The black dashed lines show the share of water that has to be condensed for compensation for the heat deficit.

Fig. 7 depicts that pressure plays an important role in the heat balance of the TCR system. As it was shown above (Fig. 6) the reaction enthalpy is strongly affecting by pressure. Increasing pressure leads to a decrease in the reaction enthalpy and as a consequence to a reduction of the heat flux in the reactor. Therefore, there is a slight heat deficit for a pressure of 10 bar.

Fig. 7 shows that the heat recuperated in the reactor is notable higher than the heat recuperated in the preheater. Fig. 8 presents a ratio between the heat recuperated in the reactor and the heat recuperated in

the preheater. As it can be seen, the recuperated heat in the reactor by the thermochemical transformation of initial ammonia is in a few times higher than the recuperated heat in the preheater by the conventional thermal recuperation due to heating.

One of the main criteria of efficiency of the thermochemical waste-heat recuperation system is the heat recovery rate determined by expression (2). However, to take into account the heat deficit, the expression (2) have to be modified as follows:

$$R = \frac{H_{rec}}{H_{exh}} = \frac{\Delta H_{298} + \Delta H_{am} - H_{deficit}}{H_{exh}} \quad (7)$$

where $H_{deficit}$ - the heat deficit.

Fig. 9 shows the heat recovery rate as a function of temperature for the pressure of 1, 5, 10, and 20 bar. The maximum HRR that can be achieved in the TCR system is about 0.8. Thus, about 80% of exhaust heat can be recovered to the fuel cycle. This HRR can be achieved at the exhaust gas temperature of 500–550°C. Fig. 9 shows the significant effect of pressure on HRR. A sharp decrease in HRR for the pressure of 1 bar is associated with heat deficit in the TCR system. For a pressure of 20 bar, a monotonic increase in HRR is observed in the temperature range of 100–500°C. Then, after reaching the maximum value of HRR at a temperature of 500°C, a monotonic decrease takes place. The monotonic decrease can be explained by the fact that in expression (2), the denominator increases faster than the numerator. This is due to the fact

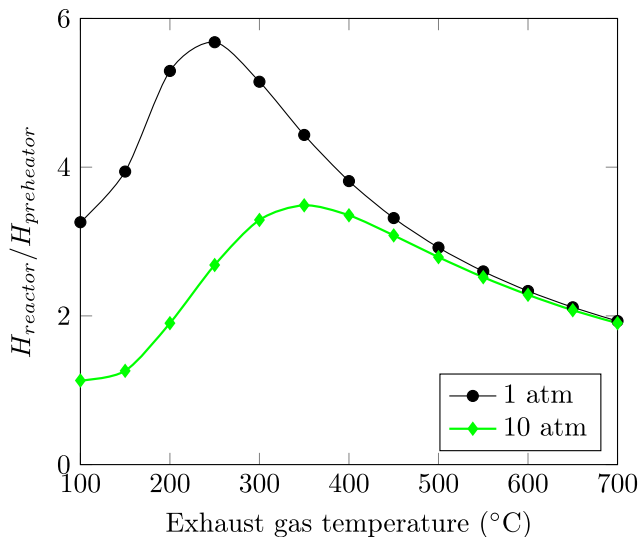


Fig. 8. A ratio between the heat recuperated in the reactor and the heat recuperated in the preheater.

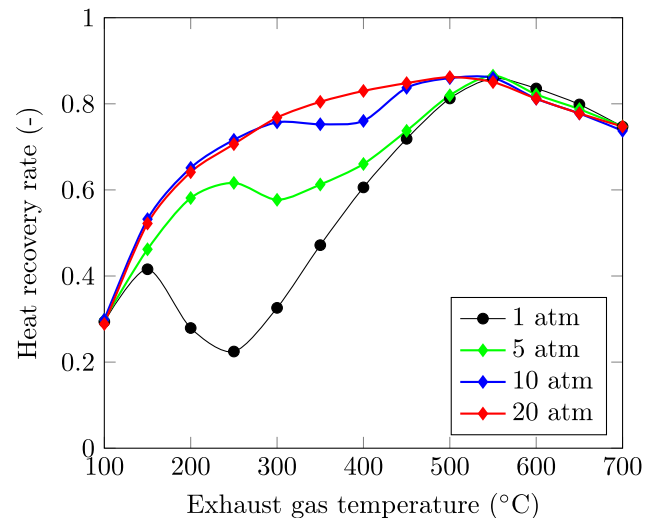


Fig. 9. The heat recovery rate as a function of temperature for pressure of 1, 5, 10 and 20 bar.

that at temperatures above 500°C, the enthalpy of the decomposition process reaches its maximum value, and heat recovery is achieved mainly due to the heating of ammonia.

Thus, the following conclusions can be drawn about the efficiency of the TCR system depending on pressure and temperature. If the exhaust gas temperature is up to 300°C, it is necessary to use ammonia with a pressure higher than 10 bar (10 bar off). At exhaust gas temperatures from 300 to 450°C, it is necessary to use ammonia with pressure above 20 bar. At temperatures above 550°C, pressure does not significantly affect the efficiency of the TCR system. In real industry, high-pressure ammonia is used. Therefore, it is advisable to use ammonia with a pressure of 20 bar and higher in the TCR systems.

Fig. 10 shows the heat released in the combustion chamber for various operating parameters of thermochemical recuperation and the heat released in the combustion chamber without TCR when only ammonia preheating is taking place. The ammonia mass flow rate is 100 kg/h Fig. 10 depicts that TCR by ammonia decomposition leads to increasing heat released in the combustion chamber by combustion. The most significant effect of TCR on the heat release takes place in the temperature range above 300°C. The ratio between the combustion heat of ammonia decomposition products and the combustion heat of pure ammonia is the heat transformation coefficient (HTC) [51]. The maximum HTC in the TCR system is about 1.16. Thus, the products of ammonia decomposition can generate 1.16 times more heat than pure ammonia.

4. Conclusion

This paper presents the results of the thermodynamic analysis of the thermochemical waste-heat recuperation system by ammonia decomposition. The thermodynamic analysis was realized via Aspen HYSYS. The results of the thermodynamic analysis were verified with the experimental results and the results of the thermodynamic analysis of other authors. Based on the thermodynamic results, the heat balance of the TCR system was calculated. The recuperated heat in the reactor by the thermochemical transformation of initial ammonia is 2–5 times higher than the recuperated heat in the preheater by the conventional thermal recuperation due to heating. For low pressure (1–10 bar) a heat deficit is appearing in the TCR system. In this paper for compensation of heat deficit, we suggested using a condenser after a preheater. To compensate for the maximum heat deficit at 300°C and 1 bar, about 13% of water from exhaust gas have to be condensed. Increasing pressure leads to a decrease in the heat deficit. The heat deficit is minimal at the pressure of 10 bar. The heat deficit leads to a decrease in the heat recovery rate. The maximum heat recovery rate (about 0.8) is in the temperature range 500–550°C. The minimum heat recovery rate is at low pressure; therefore, it is advisable to use ammonia with a pressure of 20 bar and higher in the TCR systems. The maximum heat transformation coefficient (the ratio between the combustion heat of ammonia decomposition products and the combustion heat of pure ammonia) in the TCR system is about 1.16. Thus, the products of ammonia decomposition can generate 1.16 times more heat than pure ammonia.

The thermodynamic analysis gave us opportunity to determine the maximum (ultimate level) of the energy efficiency for the thermochemical waste-heat recuperation systems by ammonia decomposition at given operation parameters such as temperature and pressure. These limit levels of the energy efficiency can be a guideline for the design of the elements of the TCR systems by ammonia decomposition. The results of this paper can be useful for application of thermochemical waste-heat recuperation in the gas turbines and other ammonia fired equipment.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

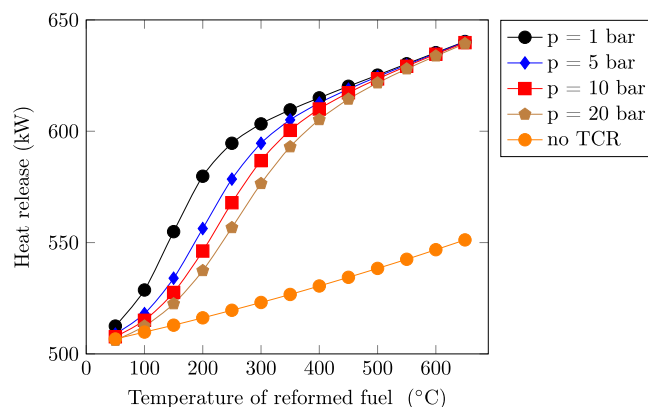


Fig. 10. The heat released in the combustion chamber as a function of temperature of the decomposition products (mass flow rate of ammonia of 100 kg/h).

the work reported in this paper.

Acknowledgment

This work is supported by the Russian Science Foundation under grant 19-19-00327.

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